

Homework 7

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Problem 1: Regression Warm Up

Part A

A 55 year old should expect a creatinine clearance of 113.71. I fit the data to a linear regression model and got the equation $\text{creatinine} = 147.81 - 0.62 * \text{age}$. By plugging in 55 for the age we get the expected creatinine clearance for a 55 year old.

Part B

The creatinine clearance rate decreases at 0.62 ml/minute per year. I fit the data to a linear regression model and determined the slope of the equation to determine the rate of change for creatinine clearance.

Part C

The expected creatinine clearance for a 40 year old is determined by the expression $147.81 - 0.62 * 40$, which evaluates to 123.01. The 40 year old therefore has a creatinine clearance 11.99 higher than expected. The expected creatinine clearance for a 60 year old is determined by the expression $147.81 - 0.62 * 60$, which evaluates to 110.61. The 60 year old therefore has a creatinine clearance 1.39 higher than expected. The 40 year old has a creatinine clearance that is healthier for their age.

Problem 2: Modeling Disease Growth

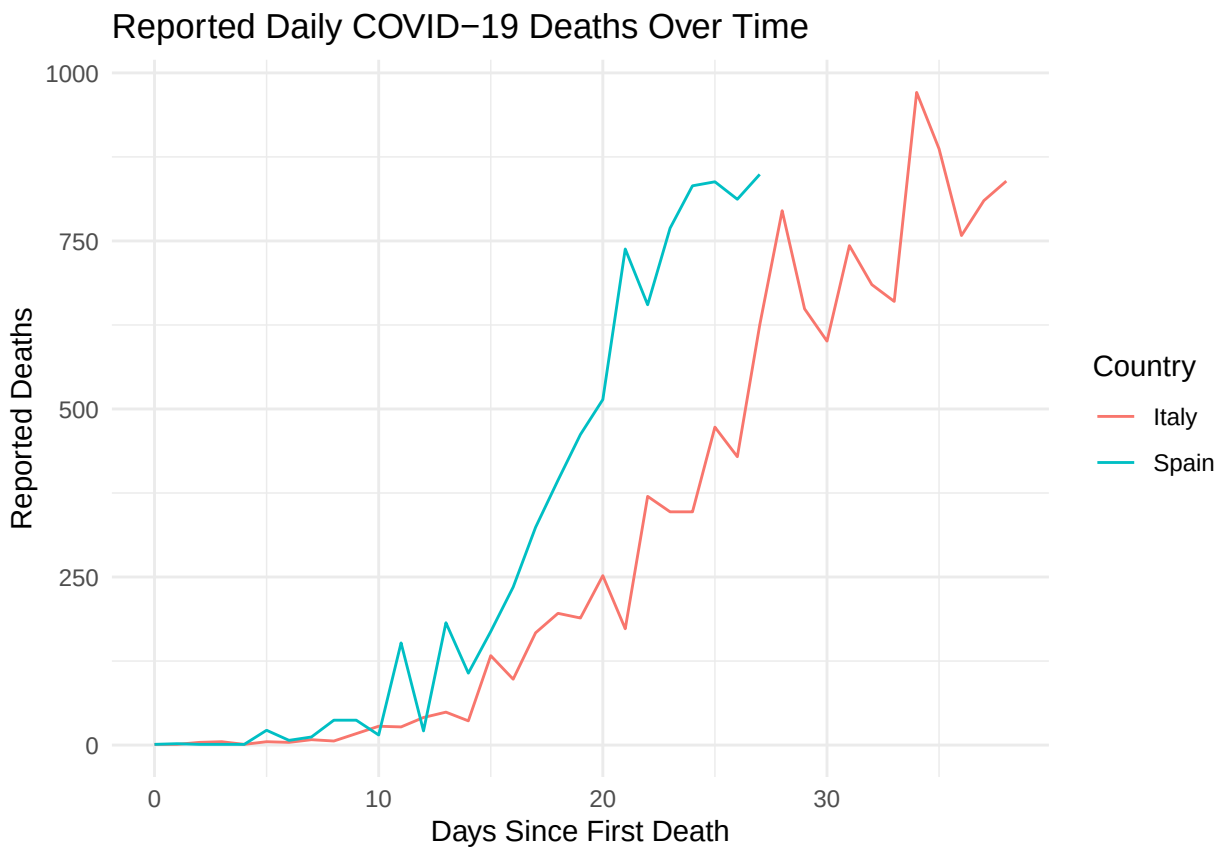
1

The estimated growth rate for Italy is 0.183 with a bootstrapped 95% confidence interval of (0.159, 0.208). The estimated doubling time for Italy is 3.8 days with a bootstrapped 95% confidence interval of (3.4, 4.4).

2

The estimated growth rate for Spain is 0.276 with a bootstrapped confidence interval of (0.235, 0.318). The estimated doubling time for Spain is 2.5 days with a bootstrapped 95% confidence interval of (2.2, 3.0).

3



Problem 3: Price Elasticity of Demand

The estimated price elasticity of demand for milk is -1.619 with a bootstrapped 95% confidence interval of (-1.772, -1.456). To estimate the elasticity, i first took the log of both sides of the equation to effectively turn it into a linear equation with Beta being the slope. I then created a linear regression model of the log of sales based on the log of prices to estimate the price elasticity before bootstrapping repeatedly from the data to create a confidence interval.